

Homework (Jalapeno)

- Q1.** The Great Pyramid of Giza took x years to build with a labor force of y workers. If the number of workers increased by 20% and the time taken decreased by 15%, which system represents the new situation?
- (A) $1.2y = 0.85x$
(B) $1.2x = 0.85y$
(C) $0.85y = 1.2x$
(D) $0.85x = 1.2y$
(E) $x = y$
- Q2.** Archimedes' Screw was used to irrigate fields. If x represents the amount of water pumped per hour and y the number of screws, which equation represents the relationship $x = 5y - 2$?
- (A) Each screw pumps 5 units of water per hour, with an initial 2-unit loss.
(B) Each screw pumps 2 units of water per hour, with an initial 5-unit gain.
(C) Each screw reduces the water pumped per hour by 2, with 5 units pumped without screws.
(D) Each screw increases the water pumped per hour by 2, with 5 units lost without screws.
(E) Each screw decreases the water pumped per hour by 5, with 2 units gained without screws.
- Q3.** A pharaoh's treasury contains x gold bars and y precious gems. If the gold bars are worth 200 each and the gems 50 each, which equation represents a total worth of 10,000?
- (A) $200x + 50y = 10000$
(B) $50x + 200y = 10000$
(C) $200x - 50y = 10000$
(D) $50x - 200y = 10000$
(E) $x + y = 10000$
- Q4.** The ancient city of Pompeii had x homes and y public fountains. If 2 public fountains served 5 homes, which system represents this relationship?
- (A) $2y = 5x$
(B) $5y = 2x$
(C) $2x = 5y$
(D) $5x = 2y$
(E) $x = y$
- Q5.** An archaeologist finds x artifacts from the Bronze Age and y from the Iron Age. If the ratio of Bronze to Iron Age artifacts is 3 : 5, which equation represents this relationship?
- (A) $x/y = 3/5$
(B) $y/x = 3/5$
(C) $x/y = 5/3$
(D) $y/x = 5/3$
(E) $x = y$
- Q6.** A historian studies the population growth of ancient civilizations. If x represents the population and y the number of years since the civilization's founding, which equation represents a population growth rate of 2% per year?
- (A) $x = 1.02^y$
(B) $x = 2y + 1$
(C) $y = 0.02x$
(D) $y = x/1.02$
(E) $x = y^2$
- Q7.** The construction of the Great Wall of China required x workers and y tons of materials. If 100 workers used 20 tons of materials per day, which system represents this relationship?
- (A) $x/100 = y/20$
(B) $x/20 = y/100$
(C) $100x = 20y$
(D) $20x = 100y$
(E) $x = y$
- Q8.** Ancient traders exchanged x units of goods for y units of currency. If 3 units of goods were equivalent to 7 units of currency, which equation represents this exchange rate?
- (A) $3x = 7y$
(B) $7x = 3y$
(C) $x/y = 3/7$
(D) $x/y = 7/3$
(E) $x = y$

Open Ended Questions (FRQ)

- Q9.** The Hanging Gardens of Babylon required a complex system of irrigation. Suppose the gardens needed x gallons of water per day and y gallons were available from local sources. If 20% of the water was lost due to evaporation and 10% was reserved for other uses, write a system of equations to represent the relationship between x and y , considering the water loss and reservation. Show your work and explain your reasoning.
- Q10.** Historians estimate that the ancient city of Angkor Wat had a population of x people and y temples. If the population grew at a rate of 2% per year and the number of temples increased by 5 each year, write a system of equations to model the relationship between x and y over a 10-year period. Solve the system and interpret the solution in context. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the historical context of each problem.
- Tip 2: Encourage students to use graphical and algebraic methods to solve systems of equations.
- Tip 3: Emphasize the importance of interpreting solutions in the context of each problem.